Hospital Data Analysis – Project Summary

Objective

The purpose of this project was to analyze a healthcare dataset using Python to uncover trends, patterns, and insights around patient demographics, hospital stays, outcomes (e.g., recovery or mortality), and admission/discharge patterns. This data-driven approach helps in understanding patient behavior, operational efficiency, and identifying areas of concern like risk factors for deceased patients.

1. Data Loading and Initial Exploration

- Loaded a CSV file (`Health_dataset.csv`) containing patient information.
- Displayed the first few rows to understand the structure.
- Counted the total number of records (patients) to get a sense of data volume.

2. Gender Distribution Analysis

- Counted and visualized the distribution of patients by gender.
- Used a bar chart with clear data labels to show the number of male vs. female patients.
- Purpose: Understand gender-based healthcare usage or potential gaps.

3. Patient Stay Duration Grouping

- Calculated the length of stay for each patient based on admission and discharge dates.
- Categorized patients into groups based on duration (e.g., 0–5 days, 6–10 days, etc.).
- Counted patients in each group to identify which durations were most common.
- Visualized this with bar charts for clarity.

4. Health Outcome Distribution

- Segregated patients into:
  - Recovered
  - Under Treatment
  - Deceased
- Counted the number of patients in each category.
- Created a doughnut chart to show the percentage share of each group.
- Purpose: Assess hospital efficiency and critical care effectiveness.

5. Mortality Risk Factor Analysis

- Filtered records where the outcome was recorded as death.
- Analyzed features like age, gender, underlying conditions to see commonalities.
- Created bar plots and possibly heatmaps to visualize correlations.
- Goal: Identify risk factors contributing to patient mortality.

6. Operational Trends – Admission & Discharge Patterns

- Analyzed which days of the week had peak admission and discharge counts.
- Further grouped data by month to understand seasonal trends.
- Line charts were used to show:
    - Monthly admissions (filtered by year)
    - Weekly and yearly patterns
- Insights: Help hospitals in planning resources and managing peak times.

- Bar chart: Gender distribution
- Bar chart: Patient count by stay duration
- Doughnut chart: Health outcomes

- Line chart: Monthly admissions
- Day-of-week trend charts: Admissions vs. Discharges
- Risk factor visuals for deceased patients

All visualizations were created using Matplotlib and Seaborn, with attention to labels, annotations, and aesthetics for clarity and readability.

- A hands-on project executed without AI assistance – relying purely on Python, data manipulation, and visualization.
- Helped uncover meaningful trends around patient behavior, hospital operations, and treatment outcomes.
- This type of analysis can aid in real-world decision-making for hospital administration and public health policy.